

POKHARA UNIVERSITY

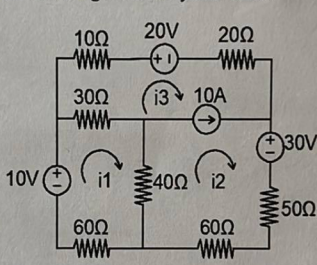
Level: Bachelor Semester: Fall Year : 2024
 Programme: BE Full Marks: 100
 Course: Basic Electrical Engineering (New) Pass Marks: 45
 Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

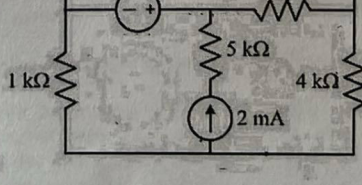
Attempt all the questions.

1. a) Explain in brief the role of electricity in modern society, Explain generation, transmission and distribution of electrical energy in context of Nepal with the help of single line diagram. 7
- b) Find i_1, i_2, i_3 current using mesh analysis method. 8



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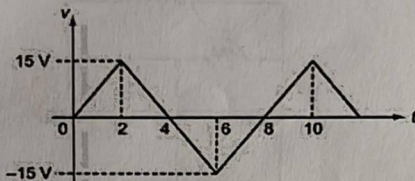
2. a) Find node voltages and current through 6 kΩ using Nodal analysis. 7



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OR

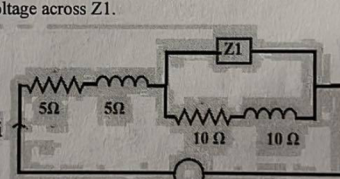
Calculate the average value, RMS (effective) value and form factor of the given waveform below.



- b) An ac voltage of $(100+j0)$ V applied to a circuit results current of $(10+j0)$ A. Find: 8

i. Impedance Z_1 of circuit.

ii. Voltage across Z_1 .



4. a) How can resonance be defined in the circuit? Also explain resonance in series RLC circuit. 7
- b) A balanced star connected load of $(8+j6)$ Ω per phase is connected to a three-phase, 50 Hz, 380 V supply. Find the phase current, line current, power factor, active power, reactive power and total volt-amps. 8
5. a) Explain the differences and analogy between magnetic circuit and electric circuit. 7
- b) A 50-kVA, 2400V/120V, 50-Hz, single-phase transformer gave the following test results: 8

OC test (HV winding open): 120 V, 9.65 A, 396 W.

SC test (LV winding shorted): 92 V, 20.8 A, 810 W.

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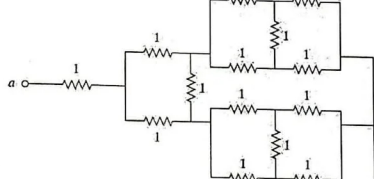
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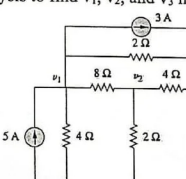
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Attempt all the questions.

1. a) Classify types of consumers based upon their electricity consumption pattern. Also draw the single line diagram of a general power system. 7
- b) Find equivalent resistance at a-b R_{ab} in the figure. Assume each element is 1Ω. 8



2. a) Using nodal analysis to find v_1, v_2 , and v_3 in the following circuit. 7

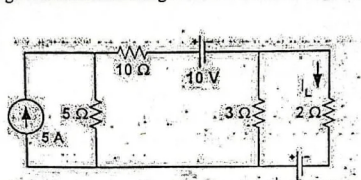


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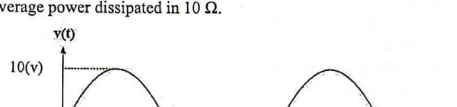
What are active and passive elements, unilateral and bilateral element? What is star delta transformation and writes its role in circuit analysis?

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- b) State Superposition Theorem power theorem. Find voltage drop through 2-ohm resistor using Norton's Theorem in given circuit. 8

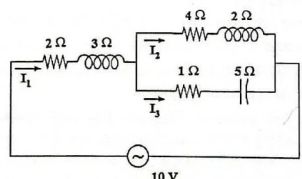


3. a) The waveform given is the half wave rectified sine wave. Find the average, RMS value and form factor. Also find the amount of average power dissipated in 10 Ω. 8



- b) For the electric circuit shown below, find: 7

- supply current and branch currents
- voltage across each branch
- Active, reactive and apparent power of entire circuit
- overall power factor
- Draw phasor diagram showing the currents and supply voltages.



4. a) With the help of series resonance curve, explain resonance in series RLC circuit. 7

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- b) Three identical coils, each of resistance 10 Ω and inductance 42 mH are connected a) in star b) in delta to a 415 V, 50 Hz, 3-φ supply. Determine: 8

i. line and phase currents

ii. Line and phase voltages

iii. the total power dissipated in each case and compare them.

5. a) Explain two wattmeter method for three phase power measurement. Also, how it is different from one wattmeter and three wattmeter method. 7

- b) A 5 kVA, 200/1000 V, 50 Hz, single phase transformer gave the following test results: 8

OC test: 200V, 1.2 A, 90 W

SC test: 50 V, 5A, 110 W

i. Calculate the parameters of the equivalent circuit referred to LV side.

ii. Calculate the output secondary voltage when delivering 3 kW at 0.8 power factor lagging, the input voltage being 200 V. Find also the percentage voltage regulation.

6. a) A 250 V DC shunt motor takes 8 A line current on no load and runs at 1000 rpm. The resistance of the field winding and armature winding are 200 Ohm and 0.8 Ohm respectively. If the full load line current is 20 A, calculate the full load speed assuming constant air gap flux. 7

- b) Explain the construction of DC generator with working principle. 8

OR

Explain the importance of transformer in power transmission. Derive the EMF equation of transformer.

7. Write short notes on: (Any two) 2×5

a) Speed Control of DC Motor

b) Color coding of resistors

c) Power in purely inductive circuit

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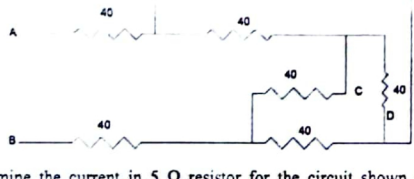
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Attempt all the questions.

1. a) With the help of single line diagram, explain generation, transmission, and distribution of electricity in Nepal. 7

OR

- b) Why do we need color coding of Resistors? Find the equivalent resistance of the following circuit. 8



2. a) Determine the current in 5 Ω resistor for the circuit shown using Superposition Theorem (Figure 22). 8

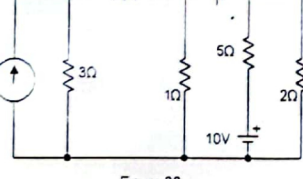
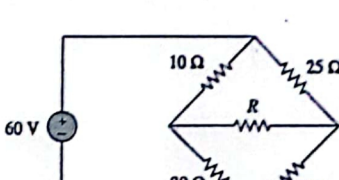


Figure 22

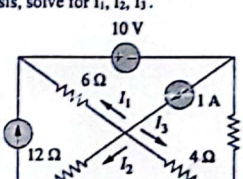
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- b) Calculate the value of R that will absorb maximum power from the circuit (shown in the figure). Also calculate the maximum power drawn by it. 7



OR

Using mesh analysis, solve for I_1, I_2, I_3 .



3. a) Determine the V_1 and V_2 using Nodal Analysis method. Also determine the current flowing through 30 Ω resistor (Figure 19) 7

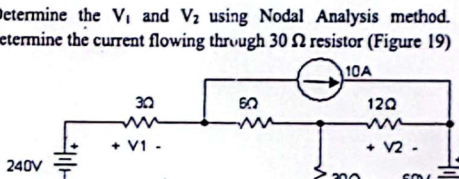
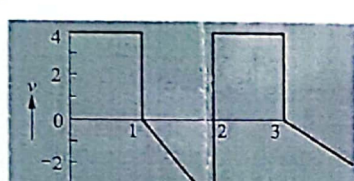


Figure 19

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- b) Find the average value, rms value of the voltage waveform given below. 8



4. a) What will be the case for resonance in series RLC circuit? Derive the resonant frequency for this case. 7

- b) A circuit have following electrical components in parallel: a) a resistor of 500 ohm, b) an inductance of 2H, c) capacitor of 10 microfarad. In this circuit 200V, 50 Hz source is applied. Calculate: 8

i. the total current drawn from the supply

ii. complex power

iii. active power

iv. reactive power

v. power factor of the circuit

5. a) The star connected stator of three phase, 50 Hz alternator supplies a balanced delta connected load. Each phase of the load consists of a coil of resistance of 15Ω and inductance 36mH and the phase voltage generated by the alternator is 231V. Calculate: 8

i. the phase and line currents

ii. the load power factor

iii. the power delivered to the load.

- b) An ideal 25 KVA transformer has 500 turns on the primary winding and 40 turns on the secondary winding. The primary winding is connected to a 3.3 KV supply. Calculate: 7

i. The secondary emf.

ii. The primary and secondary currents on full load.

iii. The maximum core flux.

6. a) Explain the operation of transformer on load and also draw phasor diagram at lagging power factor. 7

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- b) A 220 Vdc shunt motor draws a current of 5 A, and runs with a speed of 1500 rpm at no-load. When the motor is loaded, the speed drops to 1400 rpm. Calculate the current drawn by the motor at loaded condition. Given that the armature winding resistance is 0.2 Ω and field winding resistance is 110 Ω. 8

7. Write short notes on: (Any two) 2×5

a) Voltage divider Rule

b) Sources of Energy

c) Operation of DC motors

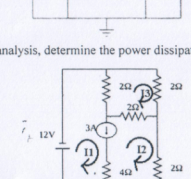
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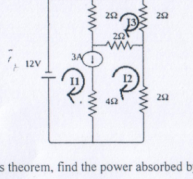
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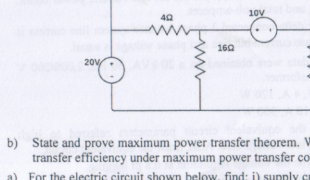
1. a) What are nodes and loop in electric circuit? For the given electric circuit, use nodal analysis to find the current through, voltage across and power consumed by the load having resistance 4 Ω. 7



- b) Using mesh analysis, determine the power dissipation in 4 Ω resistor. 8

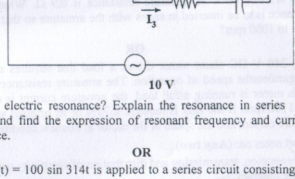


2. a) Using Thevenin's theorem, find the power absorbed by 12Ω resistor. 8



- b) State and prove maximum power transfer theorem. What is the power transfer efficiency under maximum power transfer conditions? 7

3. a) For the electric circuit shown below, find: i) supply current and branch currents ii) voltage across each branch iii) Active, reactive and apparent power of entire circuit iv) overall power factor v) Draw phasor diagram showing the currents and supply voltages. 7



- b) What is electric resonance? Explain the resonance in series RLC circuit and find the expression of resonant frequency and current at resonance. 8

OR

A voltage $v(t) = 100 \sin 314t$ is applied to a series circuit consisting of 10 Ω resistance, 0.0318 H inductance, and a capacitor of 63.6 μF. Calculate the following: 2

i. Expression for $i(t)$

ii. Power factor

iii. Active, reactive, and apparent power

iv. Also draw the phasor diagram.

4. a) A balanced star-connected load of $(8+j6)$ Ω per phase is connected to a balanced three-phase 400 V supply. Find the line current, power factor, reactive power, and total volt-amperes. 8

- b) Prove that for delta connected 3 phase balance system line current is $\sqrt{3}$ times of phase current but line and phase voltage is equal. 7

5. a) The following data were obtained on a 20 kVA, 50 Hz, 2000/200 V distribution transformer. 10

OC Test: 200 V, 4 A, 120 W

SC Test: 60 V, 10 A, 300 W

- Determine the equivalent circuit parameters referred to high voltage and low voltage side. Also, draw the equivalent circuit.
- Determine the voltage regulation at 0.8 p.f. lagging.
- Calculate the efficiency at full load condition at 0.8 p.f. lagging.

- An inductive coil of resistance 10 Ohm and inductance 0.1 H is connected in parallel with a 150 μF capacitor to a variable frequency, 200 V supply. Find the resonant frequency at which the total current taken from the supply is in phase with the supply voltage. Also, find the value of this current.

6. a) What is transformer? Explain two major application areas of transformer. Find the EMF equation of transformer. 7

- b) A 220 V dc shunt motor runs with 1500 rpm with an armature current of 60 A. The value of armature resistance is 0.9 Ω. What value of resistance is to be inserted in series with the armature so that the speed drops to 1000 rpm? OR 8

7. Write short notes on: (Any two) 2×5

- Generation, transmission and distribution of electricity using single line diagram
- Working principle and uses of three phase induction motor
- Power factor and its significance